

Project Title

SynBioRem AI

AI-Assisted Synthetic Microbe Recommendation and Interactive Simulation Platform for Pollution Remediation

This project aligns most strongly with the hackathon's **Synthetic Biology** track and **Bioremediation** track, since it focuses on engineered biological solutions and using biological systems to restore environmental balance.

1. Final project positioning

What your project is

SynBioRem AI is an **AI-driven recommendation system** that suggests the most suitable synthetic microbe design for a given pollutant and environmental condition.

It is supported by a **hands-on interactive simulation layer** where users can change pH, temperature, environment, and pollutant concentration to see how recommendations and remediation performance change.

So the correct emphasis is:

- **Primary focus:** recommendation
- **Secondary focus:** interactive simulation

2. Core problem statement

Environmental pollutants such as heavy metals, textile dyes, hydrocarbons, and pesticides are difficult to remediate efficiently because biological solutions depend on many variables:

- pollutant type
- environmental conditions
- organism suitability
- safety constraints
- deployment feasibility
- cost and scalability

Today, selecting the best biological remediation strategy often relies on slow trial-and-error methods. SynBioRem AI addresses this gap with an intelligent platform that recommends candidate microbes, proposes synthetic design blueprints, and lets users interactively test different conditions.

3. Final scope of the hackathon build

Pollutants covered

You should keep scope focused on 4 pollutants:

- Lead
- Textile dye
- Oil / hydrocarbons
- Pesticides

Candidate microbes

Use 5 candidate chassis/options:

- *Pseudomonas putida*
- *Bacillus subtilis*
- *E. coli* (lab chassis concept only)
- Microalgae
- Microbial consortium

Supported environments

- Wastewater
- Soil
- Agricultural runoff
- Controlled bioreactor

Final outputs

Your system should produce:

- Top 3 recommended microbes
- Recommendation score breakdown

- Synthetic blueprint card
- Safety score
- Cost score
- Scalability score
- Deployment recommendation
- Interactive remediation simulation graph

4. High-level system idea

The user enters a pollution scenario.

The platform analyzes it.

The recommendation engine ranks the best biological candidates.

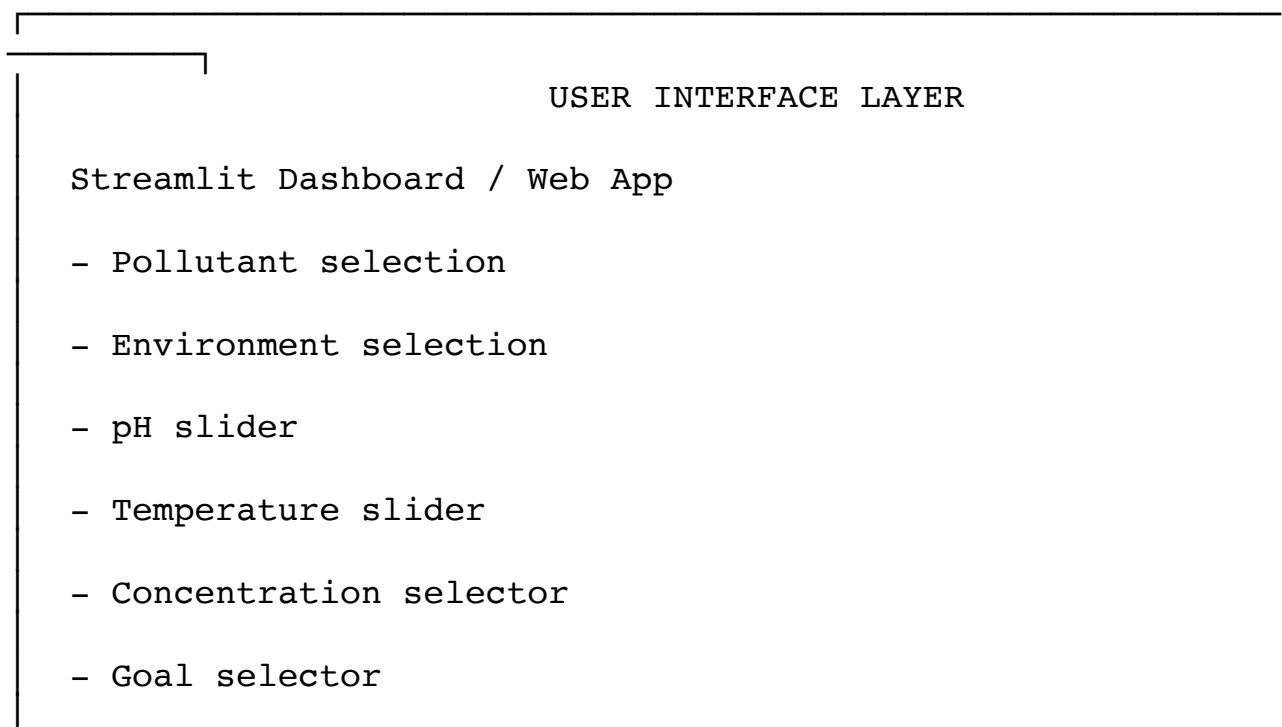
Then the simulation layer lets the user explore how changes in real-world conditions affect expected remediation performance and safety.

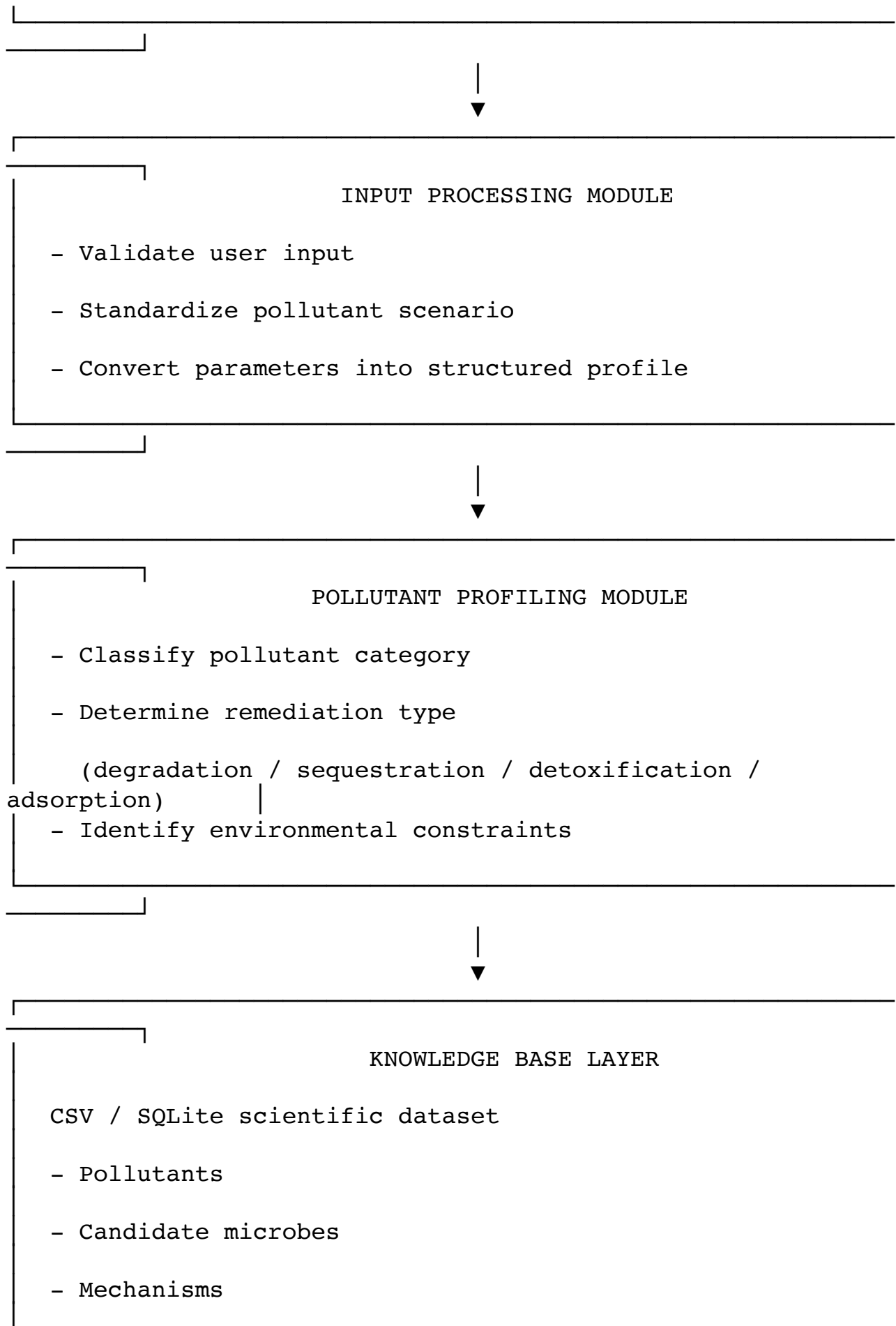
That makes the project feel like:

a scientific design assistant + interactive experimentation tool

5. Full architecture diagram

Here is a clean architecture diagram you can use in your report, PPT, or poster.





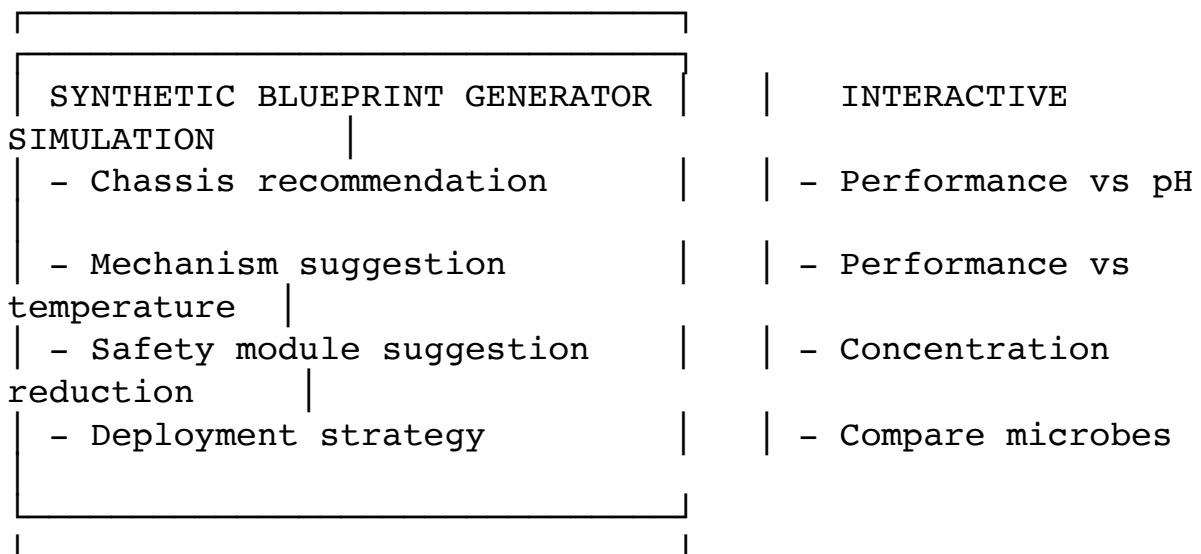
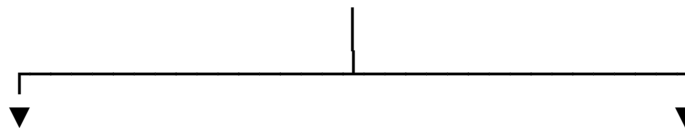
- Environmental ranges
- Safety notes
- Deployment options
- Cost and scalability values



AI RECOMMENDATION ENGINE

- Rule-based filtering
- Weighted scoring
- Candidate ranking

Output: Top 3 recommended microbes



SAFETY, FEASIBILITY & SCORING LAYER

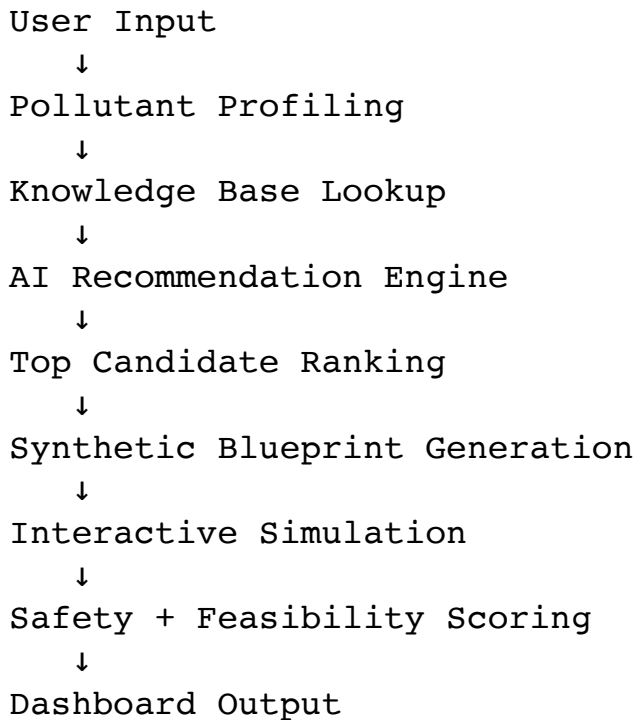
- Safety score
- Cost score
- Scalability score
- Deployment suitability
- Final recommendation score



VISUALIZATION / OUTPUT LAYER

- Ranked candidate list
- Blueprint card
- Score charts
- Simulation graphs
- Comparison dashboard
- Final remediation recommendation

6. Simplified workflow diagram



7. Module-by-module explanation

Module 1: User Input Layer

This is the dashboard section where the user interacts with the system.

Inputs

- pollutant type
- environment
- pH
- temperature
- concentration
- remediation goal

Why it matters

This makes the simulation hands-on. The user is not just watching outputs; they are actively designing a scenario.

Module 2: Input Processing Module

This converts raw user selections into a structured input format.

Example

If user selects:

- pollutant = oil
- environment = soil
- pH = 7
- temperature = 30

Then the system creates a clean scenario object for downstream modules.

Module 3: Pollutant Profiling Module

This identifies the biological logic required.

Example logic

- Lead → heavy metal → sequestration / binding
- Textile dye → organic pollutant → degradation
- Oil → hydrocarbon → hydrocarbon degradation
- Pesticide → organic pollutant → detoxification

This module decides what kind of mechanism is biologically appropriate.

Module 4: Knowledge Base Layer

This is your curated scientific dataset.

Files

- pollutants.csv
- microbes.csv
- knowledge_base.csv

What it stores

- pollutant-microbe mappings
- mechanism types
- pH and temperature compatibility
- safety risk
- deployment type
- cost score
- scalability score

This is the scientific backbone of the project.

Module 5: AI Recommendation Engine

This is the heart of the project.

Core job

Take the pollutant profile and rank the best candidate microbes.

Method

Use:

- rule-based filtering
- weighted scoring

Recommended scoring dimensions

- remediation match
- environment compatibility
- safety
- cost
- scalability

Example final score

```
Final Score =  
0.35 × Remediation Match  
+ 0.20 × Environment Compatibility  
+ 0.20 × Safety  
+ 0.10 × Cost
```

+ 0.15 × Scalability

Why this is the primary emphasis

Because this is the decision-making intelligence of the system.

Module 6: Synthetic Blueprint Generator

This produces the “designed solution” for the top candidate.

Output example

- Chassis organism: *Pseudomonas putida*
- Target pollutant: hydrocarbons
- Functional module: hydrocarbon degradation pathway
- Safety module: nutrient dependency kill switch
- Deployment: open soil or controlled bioreactor
- Suitability: high

This is what makes the project feel like synthetic biology, not just a recommendation app.

Module 7: Interactive Simulation Layer

This is the hands-on layer you wanted.

What users can do

Users can change:

- pH
- temperature
- concentration
- environment

What updates dynamically

- recommendation ranking
- expected remediation efficiency
- safety score

- deployment suitability

What should be shown

- pollutant concentration vs time graph
- microbe comparison chart
- effect of pH / temperature on efficiency
- side-by-side candidate comparison

Why this matters

It makes the demo feel alive, experimental, and scientific.

Module 8: Safety and Feasibility Layer

This ensures the system is practical and responsible.

Safety checks

- open-environment risk
- containment need
- biosafety caution
- deployment appropriateness

Feasibility checks

- cost
- scalability
- robustness
- deployment ease

This gives your outputs real-world credibility.

Module 9: Dashboard / Visualization Layer

This is what judges will see.

Main sections

- scenario input panel

- top 3 recommendations
- blueprint card
- score chart
- simulation graph
- comparison view

This layer should feel clean, visual, and convincing.

8. Detailed roadmap by phase

Phase 0 — Project framing and positioning

Goal

Lock the project identity and scope.

Tasks

- finalize title
- define one-line pitch
- define pollutants
- define microbes
- define modules
- define what recommendation means
- define what simulation means

Deliverables

- title
- abstract
- architecture draft
- finalized scope

Final output of this phase

A clear project definition:

SynBioRem AI is an AI-assisted recommendation platform for synthetic microbe design, supported by interactive simulation for pollution remediation.

Phase 1 — Scientific knowledge curation

Goal

Build your foundational dataset.

Tasks

- create pollutant list
- create candidate microbe list
- assign mechanisms
- assign pH and temperature ranges
- assign safety notes
- assign deployment types
- assign cost and scalability scores

Deliverables

- `pollutants.csv`
- `microbes.csv`
- `knowledge_base.csv`

Success criteria

You should be able to load all files and query them correctly.

Phase 2 — Recommendation engine design

Goal

Build the system that ranks microbes.

Tasks

- define scoring logic
- implement rule-based filters

- implement weighted scoring
- test sample scenarios
- produce top 3 ranked microbes

Deliverables

- `recommender.py`
- ranking logic
- tested outputs

Example scenario

Input:

- pollutant = textile dye
- environment = wastewater
- pH = 8
- temperature = 30

Output:

1. *Bacillus subtilis*
2. *Pseudomonas putida*
3. Microalgae

Phase 3 — Synthetic blueprint generator

Goal

Generate a scientific design card for top candidates.

Tasks

- define blueprint template
- map mechanisms to synthetic modules
- add safety logic
- add deployment suggestions
- produce card output

Deliverables

- `blueprint_generator.py`
- JSON/card-style structured outputs

Example output

- chassis: *Bacillus subtilis*
- mechanism: enzymatic degradation
- safety: low
- deployment: treatment plant

Phase 4 — Hands-on simulation engine

Goal

Create an interactive experimental layer.

Tasks

- define simulation formula
- model efficiency changes with pH and temperature
- model concentration reduction over time
- implement comparison mode
- make graphs dynamic

Deliverables

- `simulation.py`
- time-series output
- efficiency curves
- candidate comparison results

Hands-on features

- user moves sliders
- graphs update instantly

- recommendation changes dynamically

Phase 5 — Scoring and feasibility engine

Goal

Add practical evaluation metrics.

Tasks

- build safety scoring logic
- build cost scoring logic
- build scalability scoring logic
- combine with recommendation score
- produce final overall score

Deliverables

- `scorer.py`
- score breakdown chart
- warnings / notes

Phase 6 — Frontend / dashboard

Goal

Build the actual demo interface.

Best tool

Use **Streamlit**

Tasks

- create sidebar inputs
- show ranked results
- show blueprint
- show charts
- show comparison mode

- style dashboard cleanly

Deliverables

- `dashboard_app.py`
- working live demo

Dashboard layout

Left side:

- controls

Center:

- top recommendations
- blueprint

Right side:

- score chart
- simulation graph
- comparison results

Phase 7 — Backend integration

Goal

Organize logic cleanly.

Tasks

- connect knowledge base loader
- connect recommendation engine
- connect simulation engine
- connect blueprint generator
- build modular codebase

Deliverables

- clean Python package structure
- reusable functions

- maintainable flow

Phase 8 — Visual assets and presentation

Goal

Turn the build into a winning pitch.

Tasks

- prepare poster
- make architecture slide
- make problem-solution slide
- make workflow slide
- make demo screenshots
- prepare 1-minute and 3-minute pitch

Deliverables

- poster
- PPT
- architecture image
- demo flow

Phase 9 — Business framing and future scope

Goal

Show long-term value.

Tasks

- define users
- define B2B use case
- define wet-lab validation future
- define platform / SaaS potential

Deliverables

- business model slide
- future roadmap slide

Strong positioning

This can later evolve into a scientific software platform for:

- research labs
- environmental agencies
- industrial remediation planning

9. Recommended project structure

```
synbiorem-ai/
├── data/
│   ├── pollutants.csv
│   ├── microbes.csv
│   └── knowledge_base.csv
├── src/
│   ├── data_loader.py
│   ├── pollutant_profiler.py
│   ├── recommender.py
│   ├── blueprint_generator.py
│   ├── simulation.py
│   ├── scorer.py
│   └── utils.py
├── app/
│   └── dashboard_app.py
├── outputs/
│   ├── charts/
│   ├── reports/
│   └── sample_cases/
├── docs/
│   ├── architecture.md
│   ├── pitch_notes.md
│   └── poster_content.md
└── requirements.txt
```

10. Recommended technology stack

Core stack

- Python
- pandas
- Streamlit
- matplotlib or plotly

Optional

- FastAPI for API separation
- SQLite if you want cleaner data querying

Why this stack

Because it is fast, clean, explainable, and perfect for hackathon delivery.

11. Suggested UI layout

SynBioRem AI		
Input Panel	Recommendation Panel	Simulation Panel
- Pollutant	- Top 3 microbes	- Efficiency
graph - pH	- Blueprint card	- Comparison
chart - Temp	- Safety score	- Trend view
- Environment	- Cost score	
- Concentrat.	- Scalability score	

12. Example end-to-end scenario

Input

- pollutant: oil
- environment: soil
- pH: 7
- temperature: 30
- concentration: high

Recommendation output

1. *Pseudomonas putida*
2. Microbial consortium
3. *Bacillus subtilis*

Blueprint output

- chassis: *Pseudomonas putida*
- mechanism: hydrocarbon degradation
- safety strategy: controlled nutrient dependency
- deployment: open soil or semi-contained remediation bed

Simulation output

- expected concentration drop over 10 days
- efficiency highest near pH 7 and 30°C
- safety score decreases in open environment compared to bioreactor

That gives you a complete, believable scientific story.

13. Timeline for implementation

Day 1

- finalize scope
- verify all CSV files
- build data loader
- build pollutant profiler

Day 2

- build recommendation engine
- test 4 pollutants
- finalize scoring logic

Day 3

- build blueprint generator
- build simulation functions
- generate charts

Day 4

- build Streamlit dashboard
- connect all modules
- test interaction

Day 5

- improve visuals
- make poster
- prepare pitch and demo

If your hackathon is shorter, compress by building recommendation first and simulation second.

14. What judges will likely like

They will likely respond best to:

- originality
- scientific seriousness
- clear decision-making logic
- interactive demo
- biosafety awareness
- future commercialization potential

15. What to avoid

Do not:

- claim real synthetic microbes were created
- overcomplicate the biology
- overuse generic AI language without showing logic
- make simulation the main story
- make the UI cluttered

16. Best final one-line pitch

Use this line everywhere:

SynBioRem AI is an AI-assisted recommendation platform that proposes synthetic microbe designs for pollutant-specific bioremediation and allows users to interactively simulate environmental performance and feasibility.

17. Exact next build order

From here, the smartest order is:

1. verify Phase 1 files
2. build `data_loader.py`
3. build `pollutant_profiler.py`
4. build `recommender.py`
5. build `blueprint_generator.py`
6. build `simulation.py`
7. build `scorer.py`
8. build `dashboard_app.py`